

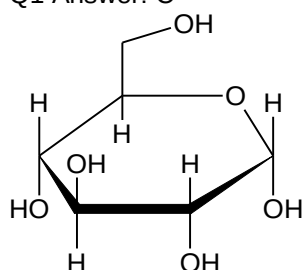
2012 9647 H2 Chemistry

Paper 1 Answer Key:

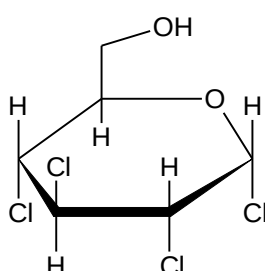
1	C	11	D	21	D	31	B
2	C	12	C	22	B	32	D
3	D	13	B	23	B	33	A
4	B	14	A	24	D	34	B
5	C	15	C	25	C	35	B
6	A	16	D	26	A	36	B
7	D	17	D	27	C	37	D
8	C	18	C	28	B	38	A
9	A	19	C	29	B	39	B
10	D	20	C	30	A	40	B

Paper 1 Worked Solutions

Q1 Answer: C



There are 4 hydroxyl groups directly attached to the ring, so 4 Cl atoms will replace them.

Molecular formula: $C_6H_8Cl_4O_2$ Empirical formula: $C_3H_4Cl_2O$ 

Q2 Answer: C

[O] $Fe \rightarrow Fe^{2+} + 2e^-$ [R] $Fe^{3+} + e^- \rightarrow Fe^{2+}$ Overall: $Fe + 2Fe^{3+} \rightarrow 3Fe^{2+}$

If 1 mol of Fe and 2 mol of Fe^{3+} react, 3 mol of Fe^{2+} form with no remaining reactants.

If equal number of moles of Fe^{3+} and Fe^{2+} must be present after the reaction, an **additional** 3 mol of Fe^{3+} must be used.

Hence, for 1 mol of Fe used, $(2 + 3 =) 5$ mol of Fe^{3+} must be used to produce 3 mol of Fe^{2+} and an excess of 3 mol of Fe^{3+} .

Q3 Answer: D

Sc: $[Ar] 3d^1 4s^2$

Only Sc does not contain an unpaired s electron or an unpaired p electron.

A: Cr $[Ar] 3d^5 4s^1$ B: Ge $[Ar] 3d^{10} 4s^2 4p^2$ C: S^- $[Ne] 3s^2 3p^5$

Q4 Answer: B

After the **first two** electrons are removed (on the graph, the two lowest IEs), there is a **large increase** in IE, indicating that the third electron is removed from an inner principal quantum shell. There are **two valence** (outermost) electrons in the element.

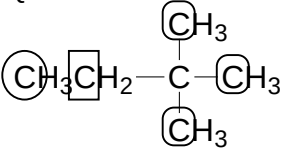
Hence, element X is in group II of the Periodic Table.

A: d-block elements have a starting series of more than two electrons (from 4s and 3d subshells) that require approximately the same amount of energy to remove.

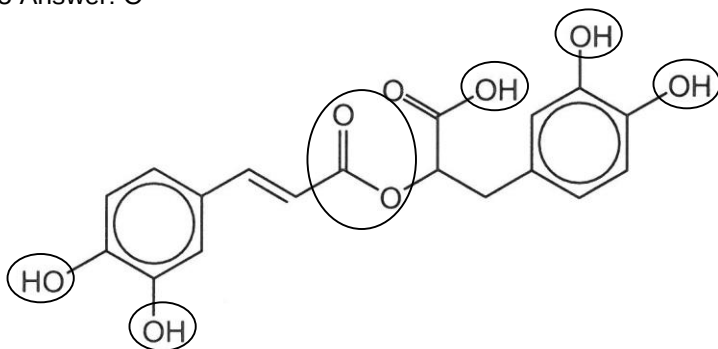
C: The first three IEs (graph should show three lowest IEs) of a group III element should gradually increase before a large increase in IE is observed.

D: An element from period 2 should have only one large increase in IE, as it has only one inner principal quantum shell.

Q5 Answer: C Every H₂O molecule has <u>two H atoms bonded to O</u> atom and <u>two lone pairs of electrons on O</u> atom. On average, each H₂O molecule can form <u>two hydrogen bonds</u>. Every HF molecule has <u>one H atom bonded to F</u> atom and <u>three lone pairs of electrons on F</u> atom. On average, each HF molecule can only form <u>one hydrogen bond</u>, limited by the single H atom. A: H and F atoms have a greater difference in electronegativity than H and O atoms, so the hydrogen bond formed between H and F atoms is stronger. However, strength of hydrogen bond is only one of two factors affecting extensiveness of hydrogen bonding. B: Number of electrons in an H₂O molecule and in an HF molecule is the same (18 electrons) hence extensiveness of van der Waals' forces is similar. D: H and F atoms have a greater difference in electronegativity than H and O atoms, so there is a greater permanent dipole in the HF molecule.	Q6 Answer: A Hydrogen bonds are formed, releasing energy, hence the reaction is exothermic ($\Delta H < 0$). The forward reaction in the equation shows a decrease in amount of gaseous particles ($\Delta n = -1$), resulting in a decrease in disorderliness in the system ($\Delta S < 0$).																								
Q7 Answer: D By Avogadro's Law (for gases, volume ratio = mole ratio), $Cl_2 \equiv 2O_2$ Electrolysis 1: $2Cl^- \rightarrow Cl_2 + 2e^-$ Electrolysis 2: $4H_2O \rightarrow 2O_2 + 8H^+ + 8e^-$ Ratio of electrons transferred in electrolysis 1 to that in electrolysis 2 = 1 : 4 Since $Q = It = n_e F$, when t is constant (5 min), $I \propto n_e$ Current used in electrolysis 2 = $4 \times$ current used in electrolysis 1 = 4I	Q8 Answer: C <table><tr><td></td><td>$2H_2(g)$</td><td>+</td><td>$CO(g)$</td><td>$\rightleftharpoons$</td><td>$CH_3OH(g)$</td></tr><tr><td>Initial amount / mol</td><td>2.0</td><td></td><td>1.0</td><td></td><td>0</td></tr><tr><td>Change in amount / mol</td><td>-x</td><td></td><td>-1/2x</td><td></td><td>+1/2x</td></tr><tr><td>Equilibrium amount / mol</td><td>$2.0 - x$</td><td></td><td>$1.0 - 1/2x$</td><td></td><td>$1/2x$</td></tr></table>		$2H_2(g)$	+	$CO(g)$	$\rightleftharpoons$	$CH_3OH(g)$	Initial amount / mol	2.0		1.0		0	Change in amount / mol	-x		-1/2x		+1/2x	Equilibrium amount / mol	$2.0 - x$		$1.0 - 1/2x$		$1/2x$
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Initial amount / mol	2.0		1.0		0																				
Change in amount / mol	-x		-1/2x		+1/2x																				
Equilibrium amount / mol	$2.0 - x$		$1.0 - 1/2x$		$1/2x$																				
Q9 Answer: A Since pyruvic acid, a weak acid, is titrated with NaOH(aq), a strong base, there will be an <u>acidic buffer</u> present before equivalence point due to a mixture of unreacted weak acid and the salt, CH₃COCO₂Na, formed from neutralisation. On the graph, this is indicated by the <u>initial sharp increase in pH followed by flattening</u> of the curve. At equivalence point, <u>pH is greater than 7</u> due to <u>salt hydrolysis</u>, which produced OH⁻ ions, as shown in the equation below. $CH_3COCO_2^- + H_2O \rightleftharpoons CH_3COCO_2H + OH^-$	Q10 Answer: D $HA \rightleftharpoons H^+ + A^-$ Due to the high [H⁺] in stomach juices (pH = 1), the dissociation of HA is suppressed as equilibrium position shifts to the left. Thus, [H⁺] is highest, followed by [HA], followed by [A⁻] (from suppressed dissociation of a weak acid).																								
Q11 Answer: D Since NaOH is the limiting reagent, Amount of HCl remaining = $0.0040 - 0.0025 = 0.0015$ mol [H⁺] remaining = $0.0015 \text{ mol dm}^{-3}$ $pH = -\lg(0.0015) = 2.82$	Q12 Answer: C Amount of uranium: 100% → 50% → 25% Amount of lead: 0% → 50% → 75% Assuming a 1:1 reaction mole ratio, it takes two half-lives to attain a 1:3 ratio of uranium (25% of reactant remaining) to lead (75% of product formed). Time taken = $2 \times 4.5 \times 10^9 = 9.0 \times 10^9$ years																								
Q13 Answer: B The rate at which a reactant is being used up is <u>not affected by its own concentration</u>. Hence, within the same amount of time at the same temperature, the same percentage (10%) of H₂O₂ will be used up if only [H₂O₂] is changed.	Q14 Answer: A The catalysed reaction occurs in <u>two steps</u>, so the graph should show <u>two "humps"</u>. <u>Enthalpy change of reaction does not change</u> with the addition of a <u>catalyst</u>, so the graph should show the <u>same energy level</u> for the (reactants and) <u>products</u> for both catalysed and uncatalysed reactions.																								

Q15 Answer: C Using knowledge of copper chemistry, a change in solution colour from blue to green may be represented by the following ligand exchange reaction. $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^- \rightleftharpoons [\text{CuCl}_4]^{2-} + 6\text{H}_2\text{O}$ In ligand exchange, oxidation number, and hence number of electrons, in copper does not change. However, different ligands result in different extents of d-orbitals splitting hence energy gap between the d-orbitals changes.	Q16 Answer: D A: Down group II, reactivity of metals increases. Hence, magnesium reacts less readily with water than barium. B: The reaction between $\text{CaO}(\text{s})$ and liquid water occurs readily. C: Magnesium reacts less readily with liquid water than with steam, since the latter occurs at a higher temperature.
Q17 Answer: D A, B: It has no large increase across the first three IEs and hence cannot be from groups I to III. C: The element has high density (take density of water at 1 g cm^{-3} as a benchmark) and high melting point and hence cannot be a non-metal with a simple molecular structure.	Q18 Answer: C Reaction between chlorine and hot $\text{NaOH}(\text{aq})$ produces NaClO_3, sodium chlorate(V).
Q19 Answer: C A: Present in the $\text{C}=\text{C}$ bond. B: Present in the $\text{C}-\text{H}$ bond next to the $\text{C}\equiv\text{C}$ bond. D: Present in the $\text{C}-\text{C}$ bond next to the $\text{C}=\text{C}$ bond.	Q20 Answer: C  There are three different C atoms that the $\text{C}\cdot$ radical can obtain a H atom from.
Q21 Answer: D Reducing a ketone without reducing alkenes and the carboxylic acid requires NaBH_4. A, B: Reduces alkenes together with ketones. C: Not a reducing agent.	Q22 Answer: B With mild oxidation, a diol forms across the $\text{C}=\text{C}$ bond in the cyclohexene ring, hence W is true and X is false. The C atoms in this $\text{C}=\text{C}$ bond become chiral as they each have four different substituents bonded to them now, hence Y is false and Z is true.
Q23 Answer: B In a nucleophilic substitution reaction, the leaving group must be δ^- to induce δ^+ on the C atom, such that C atom can attract the negatively-charged anionic nucleophile.	Q24 Answer: D Ultraviolet light in the upper atmosphere can $\text{C}-\text{Cl}$ bonds and any bonds that are weaker to undergo homolytic fission, forming free radicals that destroy ozone. Butane consists of only $\text{C}-\text{C}$ and $\text{C}-\text{H}$ bonds, which cannot undergo homolytic fission under ultraviolet light. A, B: The $\text{C}-\text{Br}$ bond is longer and weaker than the $\text{C}-\text{Cl}$ bond and can undergo homolytic fission to form $\text{Br}\cdot$. C: $\text{C}-\text{Cl}$ bond can undergo homolytic fission to form $\text{Cl}\cdot$.

Q25 Answer: C

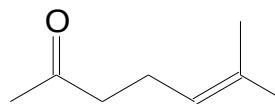


For 1 mol of rosmarinic acid:

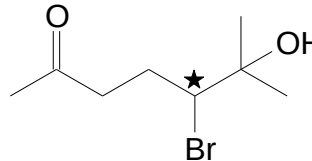
- **4 mol of phenol** and **1 mol of carboxylic acid** will undergo neutralisation with 5 mol of NaOH(aq).
- **1 mol of ester** will undergo basic hydrolysis with 1 mol of NaOH(aq).

Q26 Answer: A

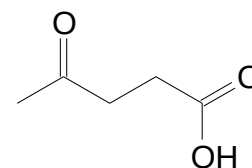
The structure of 6-methylhept-5-en-2-one is as follows.



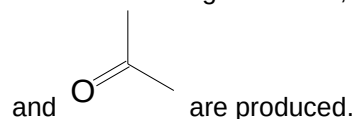
Addition of Br₂(aq) produces



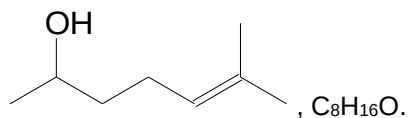
, which is chiral.



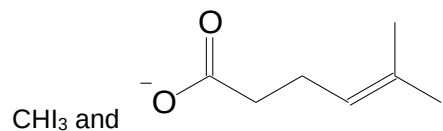
B: Under strong oxidation,



C: Only the ketone undergoes reduction with methanolic NaBH₄, which produces



D: Mild oxidation with alkaline aqueous I₂ produces



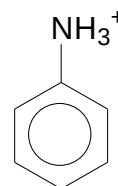
Q27 Answer: C

The **secondary structures** of proteins (α -helix and β -pleated sheets) are stabilised by **hydrogen bonding only**.

Other R group interactions (disulfide bridges, van der Waals' forces and ionic bonds) are only present in the tertiary and quaternary structures of proteins.

Q28 Answer: B

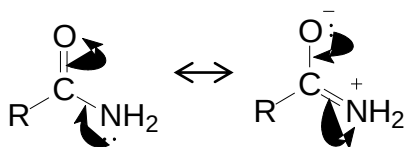
After reaction with tin and concentrated hydrochloric



acid, is produced due to the acidic medium. Hence sodium hydroxide is required to **neutralise the product** and liberate phenylamine.

Q29 Answer: B

Amides are able to display the following resonance structures.



A: Amino acids, not amides, form zwitterions.

C: Basicity involves the formation of an N-H bond through the acceptance of H^+ , not the breaking of the C-N bond.

D: The N-H bonds in amides cannot be broken as the resulting negative charge on the N atom is not stable, hence amides cannot donate H^+ and are not acidic.

Q31 Answer: B

1: Of the four valence electrons in each C atom in graphite, **one valence electron is delocalised**.

2: The other three valence electrons are involved in **covalent bonds with three other C atoms**.

3: Since C atom has **four valence electrons** in total, its valency is 4.

Q33 Answer: A

1: When **[1-bromobutane] doubles** from 0.1 mol dm^{-3} to 0.2 mol dm^{-3} while $[HS^-]$ is held constant, **initial rate of reaction also doubles** from $1.5 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$ to $3.0 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$. Hence, reaction is first-order with respect to 1-bromobutane.

2: When **[HS⁻] doubles** from 0.1 mol dm^{-3} to 0.2 mol dm^{-3} while [1-bromobutane] is held constant, **initial rate of reaction also doubles** from $1.5 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$ to $3.0 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$. Reaction is first-order with respect to HS^- . Hence, rate-determining step of the reaction involves both 1-bromobutane and HS^- .

3: From the first experiment,

$$\text{Rate constant} = (1.5 \times 10^{-5}) \div (0.1)(0.1) = 1.5 \times 10^{-3} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

Q35 Answer: B

1: Cl^- (O.N. = -1) is oxidised to Cl_2 (O.N. = 0).

2: Cl_2 (O.N. = 0) is reduced to HCl (O.N. = -1).

3: Cl^- (O.N. = -1) becomes HCl (O.N. = -1).

Q30 Answer: A

$-NH_2$ is strongly-activating and directs Br substituents to the 2, 4 and 6 positions on benzene. However, since **position 2 is occupied** by a methyl group, **only positions 4 and 6 will be substituted by Br**.

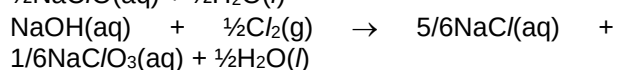
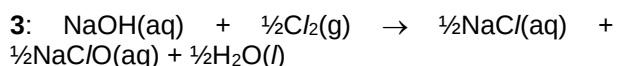
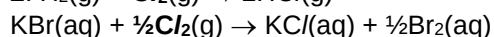
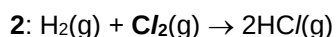
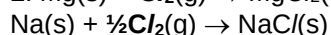
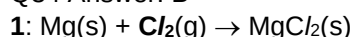
Q32 Answer: D

1: Decreasing the time taken **decreases the amount of electrons** transferred, which **decreases the amount of Cu^{2+} reduced**. A smaller mass of copper will be deposited on the cathode.

2: Increasing the concentration of the solution **does not affect the rate at which the electrons are transferred**. The rate of reduction of Cu^{2+} , and hence the mass of copper deposited, will not be affected.

3: The nature of the anion present **affects the reaction that takes place at the anode**, not the cathode.

Q34 Answer: B



Q36 Answer: B

1: Transition elements have variable oxidation states due to their ability to lose electrons from both 4s and 3d orbitals. **Calcium can only lose two electrons** to form Ca^{2+} .

2: **Ca^{2+} cannot undergo reduction** as seen from its highly negative $E^\ominus$ value of -2.87 V. Hence, it is unable to act as an oxidising agent.

3: **All cations have radii that are less than their corresponding atomic radii**. In forming cations, nuclear charge remains the same but there are fewer electrons, hence electrostatic forces of attraction for each electrons by the nucleus increases.

Q37 Answer: D

1: As the reaction proceeds, propanol (an alcohol) is formed. Hence, **alcoholic NaOH** would be present to cause the **elimination of HBr** from the halogenalkane $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ to form $\text{CH}_3\text{CH}=\text{CH}_2$.

2: Propanol is a **neutral** product and does not undergo neutralisation with NaOH.

3: It is **not possible for a hydroxyl group to be introduced onto position 2** in the carbon skeleton, since Br in the reactant is on position 1.

Q39 Answer: B

1: There is **p-orbital overlap** between the Br atom and the two C atoms, allowing the electrons of Br atom to delocalise into the π bond.

2: The π bond is **electron-rich** and repels electron-rich nucleophiles.

3: The π bond **restricts C=C bond rotation**, not the C-Br bond.

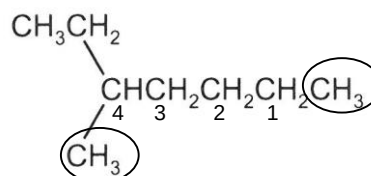
Q38 Answer: A

1: **Free radical substitution** can occur with any of the C-H bonds.

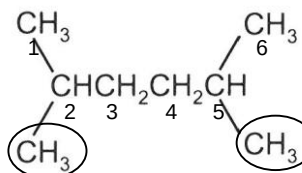
2: **Nucleophilic substitution** can occur where each of the three alcohol groups are replaced by Cl.

3: **Reduction** of ketone to form an alcohol.

Q40 Answer: B



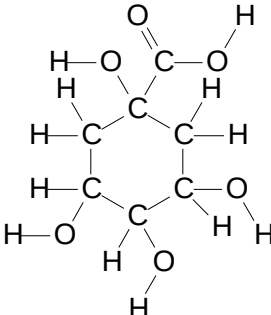
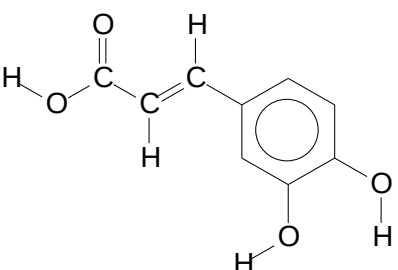
1: C atoms 1 and 6 can form a bond to produce the benzene ring, with two methyl group substituents on positions 1 and 4.

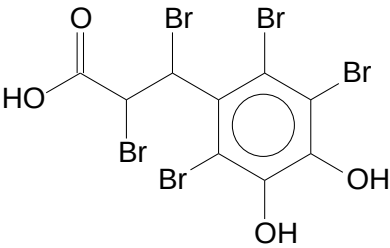
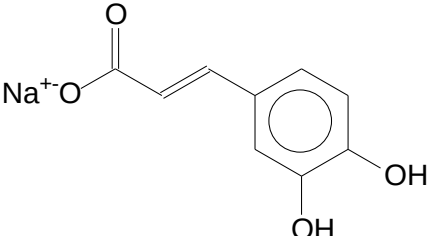
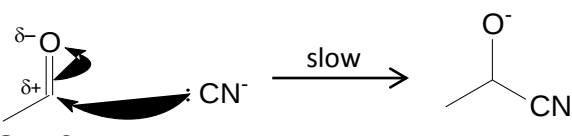
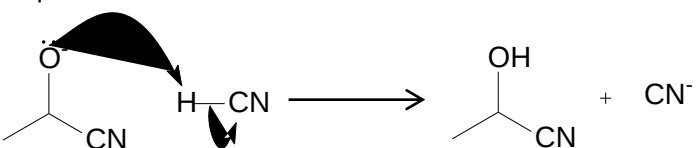
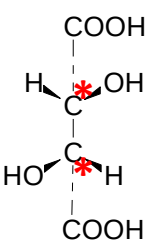
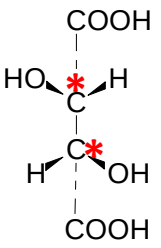


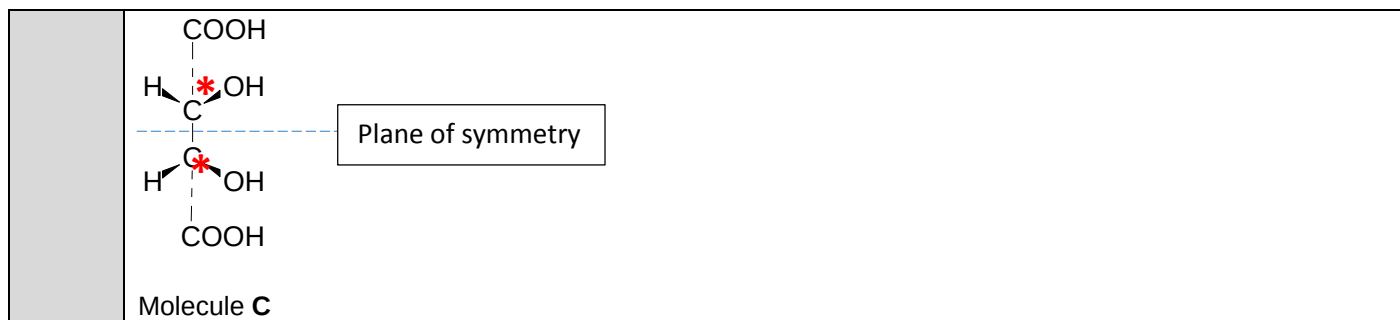
2: C atoms 1 and 6 can form a bond to produce the benzene ring, with two methyl group substituents on positions 1 and 4.

3: Not possible to form a bond between any two C atoms to result in two methyl substituents on positions 1 and 4.

Paper 2 Worked Solutions

2(a)(i)	Hydrocarbons / alkanes
2(a)(ii)	In the car engine, nitrogen reacts with limited O ₂ to form NO gas at high temperature.
2(a)(iii)	Sulfur comes from proteins or amino acids that are present in the hydrocarbon fuel
2(b)(i)	$\Delta H_{\text{rxn1}}^\ominus = \Delta H_f^\ominus(\text{SO}_3) - \Delta H_f^\ominus(\text{SO}_2) = -396 - (-297) = -99.0 \text{ kJ mol}^{-1}$
2(b)(ii)	$\Delta G_{\text{rxn1}}^\ominus = \Delta H_{\text{rxn1}}^\ominus - T\Delta S_{\text{rxn1}}^\ominus$ $\Delta G_{\text{rxn1}}^\ominus = (-99) - 298(-0.0935) = -71.1 \text{ kJ mol}^{-1}$
2(c)(i)	$K_c = 10^{\frac{\Delta G^\ominus}{-2.303RT}} = 10^{\frac{-71140}{-2.303(8.31)(298)}} = 2.98 \times 10^{12} \text{ mol}^{-1/2} \text{ dm}^{3/2}$
2(c)(ii)	Since $K_c > 1$, equilibrium position in $\text{SO}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{SO}_3(\text{g})$ lies more to the right, indicating that at 298 K, most / all of the $\text{SO}_2(\text{g})$ is oxidised to $\text{SO}_3(\text{g})$.
2(c)(iii)	The activation energy for the oxidation reaction is too high, causing $\text{SO}_2(\text{g})$ to be kinetically stable relative to $\text{SO}_3(\text{g})$.
2(c)(iv)	Addition of a catalyst provides an alternative pathway of lower activation energy for the oxidation reaction to occur. Thus speeding up the rate of reaction.
3(a)(i)	$K_{\text{sp}} = [\text{Pb}^{2+}][\text{Cl}^-]^2$ Units: $\text{mol}^3 \text{ dm}^{-9}$
3(a)(ii)	Let $s \text{ mol dm}^{-3}$ be the solubility of PbCl_2 . $s = \frac{4.7}{278.0} = 0.01691 \text{ mol dm}^{-3}$ $K_{\text{sp}} = s(2s)^2 = 4s^3 = 4(0.01691)^3 = 1.9 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$ (2s.f.) Note: the final answer must be presented in 2 s.f. required in the question.
3(b)(i)	Immediately, white precipitate After one hour in daylight, black precipitate is seen.
3(b)(ii)	Addition of aqueous ammonia would cause the white precipitate to dissolve, giving a colourless solution (diammine silver (I) complex form) $\text{AgCl}(\text{s}) + 2\text{NH}_3(\text{aq}) \rightarrow [\text{Ag}(\text{NH}_3)_2]^+(\text{aq}) + \text{Cl}^-(\text{aq})$
3(b)(iii)	A ligand exchange reaction occurs, forming a lead(IV) complex that is soluble in water. $\text{PbCl}_2 + 2\text{Cl}^- \rightarrow [\text{PbCl}_4]^{2-}$
3(c)	$[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 4\text{Cl}^- \rightleftharpoons [\text{CuCl}_4]^{2-} + 6\text{H}_2\text{O}$ <u>Ligand exchange</u> between H_2O molecules and Cl^- ions took place. When concentrated HCl is added, $[\text{Cl}^-]$ increases. By Le Chatelier's principle, equilibrium position shifts to the right in order to reduce $[\text{Cl}^-]$, using up blue $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ ions and forming more yellow $[\text{CuCl}_4]^{2-}$ ions.
3(d)(i)	Ionic product of each metal sulfide = $[\text{M}^{2+}][\text{S}^{2-}]$ In acidic solution, high $[\text{H}^+]$ causes the equilibrium position in $\text{HS}^-(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{S}^{2-}(\text{aq})$ to shift to the left to reduce $[\text{H}^+]$, at the same time reducing the $[\text{S}^{2-}]$ such that ionic product of CuS exceeds its K_{sp} value but not for NiS and ZnS . Since CuS has the lowest K_{sp} value, it will be the easiest to precipitate. Hence, precipitation of CuS occurs but not for NiS and ZnS .
3d(ii)	In alkaline solution, OH^- neutralises the H^+ present, causing $[\text{H}^+]$ to fall. The equilibrium position in $\text{HS}^-(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{S}^{2-}(\text{aq})$ shifts to the right to increase $[\text{H}^+]$, at the same time increasing the $[\text{S}^{2-}]$ such that ionic product of NiS exceeds its K_{sp} value, causing the precipitation of NiS . NiS has the highest K_{sp} value, it will be the hardest to precipitate.
3d(iii)	In neutral solution, $[\text{H}^+]$ ions from dissociation of water is low enough such that the equilibrium position in $\text{HS}^-(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{S}^{2-}(\text{aq})$ lies sufficiently to the right and $[\text{S}^{2-}]$ is high enough to cause ionic product of ZnS to exceed its K_{sp} value, causing precipitation of ZnS .
4(a)(i)	2° alcohol, 3° alcohol, phenol, ester, carboxylic acid
4(a)(ii)	4
4(b)	 non-aromatic compound  aromatic compound R

4(c)	 Note: the 2 phenolic –OH group activates the benzene ring, causing at least 2 Br atoms to be substituted in the ring.
4(d)(i)	
4(d)(ii)	The carboxylic acid functional group in R is a stronger acid than the phenol functional group. Therefore only the carboxylic acid functional group reacts with NaHCO₃.
5(a)(i)	Reagent: HCN(aq) Conditions: NaCN(aq) or NaOH(aq) catalyst, 10–20°C Type of reaction: Nucleophilic addition
5(a)(ii)	Step 1: $\text{HCN} \rightleftharpoons \text{H}^+ + \text{CN}^-$ $\text{NaCN} \rightarrow \text{Na}^+ + \text{CN}^-$ Step 2:  Step 3: 
5(b)(i)	All molecules present are of same chirality so there is only one enantiomer present.
5(b)(ii)	The nucleophile CN^- has equal probabilities of attacking the trigonal planar carbonyl carbon atom in ethanal from the top or bottom of the plane, causing an optically inactive racemic mixture of optical isomers to be formed.
5(c)	Molecule A contains two chiral carbon atoms with no plane of symmetry. Molecule B is the optical isomer of molecule A and hence rotate plane polarised light in the opposite direction.  Molecule A  Molecule B Molecule C contains two chiral carbon atoms but there is a plane of symmetry and hence is optically inactive.

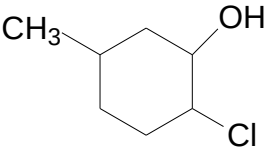
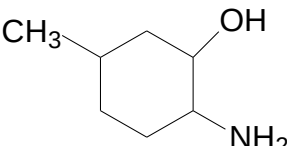
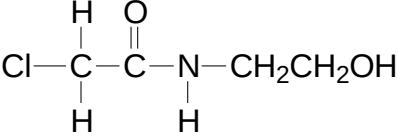
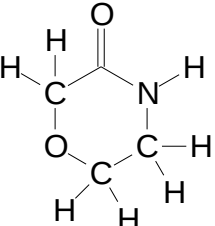
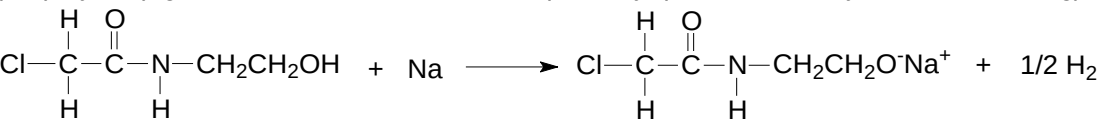
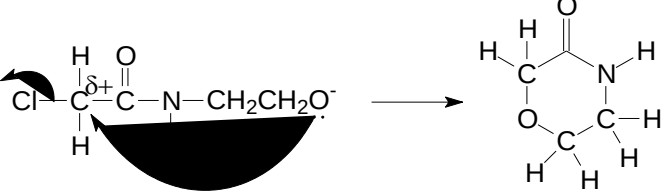


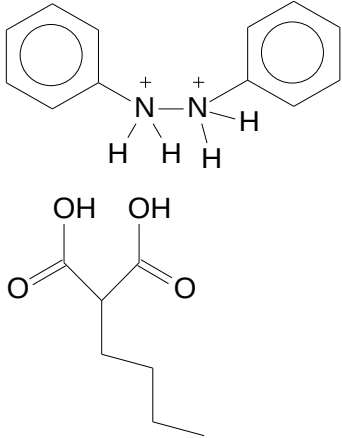
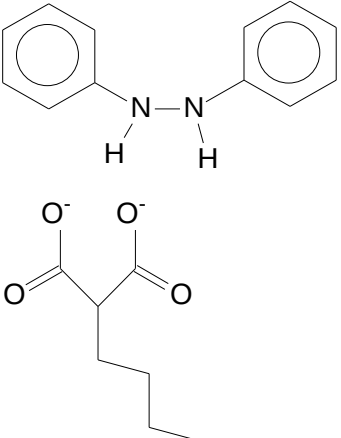
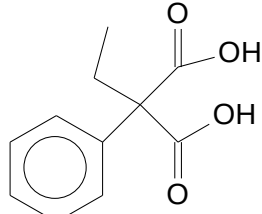
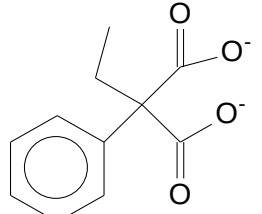
Paper 3 solutions

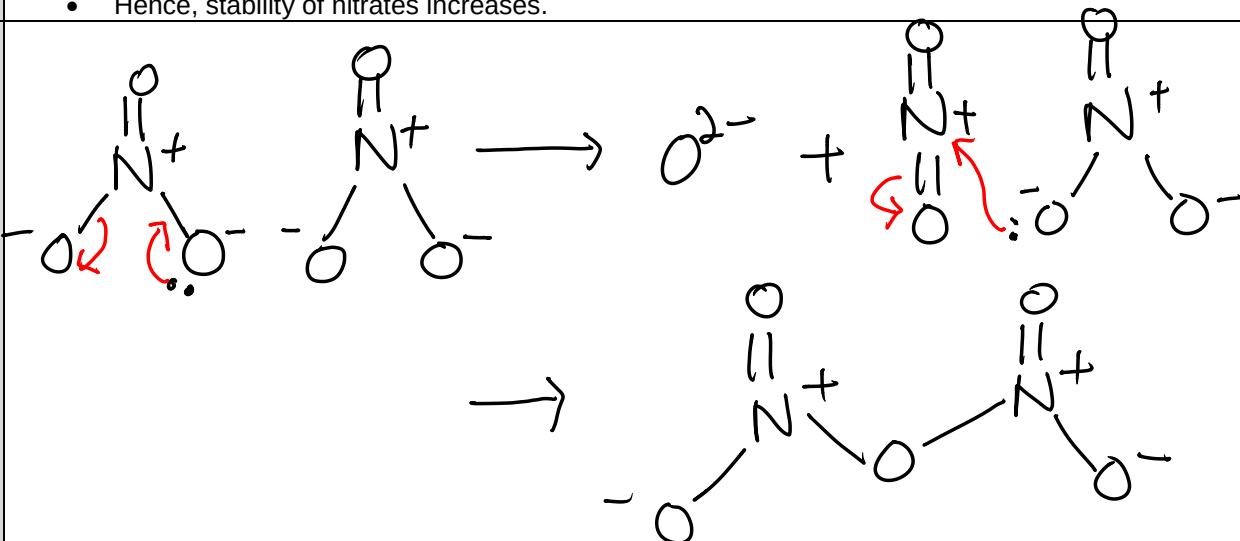
1(a)	Basic oxide: <u>MgO</u> Amphoteric oxide: <u>Al₂O₃</u> Acidic oxide: <u>P₄O₁₀</u> MgO and P₄O₁₀ undergo <u>acid-base</u> reaction to give <u>salt and water</u>. MgO + 2HCl → MgCl₂ + H₂O MgO does not react with alkalis. P₄O₁₀ + 12NaOH → 4Na₃PO₄ + 6H₂O P₄O₁₀ does not react with acids. Al₂O₃ + 6HCl → 2AlCl₃ + 3H₂O Al₂O₃ + 2NaOH + 3H₂O → 2Na[Al(OH)₄] <u>Al₂O₃ reacts with both acids and alkalis.</u> <u>Aluminium oxide</u> is an <u>ionic compound with marked degree of covalent character</u>. This is due to the <u>high charge density of the Al³⁺</u> thus it has <u>high polarising power to distort the oxide (O²⁻) electron cloud</u>. This results in partial covalent characteristic in Al₂O₃ which allow it to behave like covalent oxide similar to P₄O₁₀.
1(b)(i)	
1(b)(ii)	Shape of AlCl ₃ → Trigonal Planar
1(b)(iii)	Al₂Cl₆ ⇌ 2AlCl₃ ΔH = +63 kJ mol⁻¹ Forward reaction is endothermic. When temperature increases, by LCP, the equilibrium position will favour the endothermic reaction where the system will absorb the additional energy. Thus equilibrium position will shift right favouring the formation of AlCl₃.
1(b)(iv)	pV = nRT (1.16 × 10⁵)(2.5 × 10⁻⁴) = (1.50/M_r)8.31(500) M_r = 214.9
1(b)(v)	x + y = 1 y = 1 - x -----(1) 214.9 = 267x + 133.5y -----(2) Sub (1) into (2) x = 0.610 y = 0.390
1(b)(vi)	Partial Pressure of Al ₂ Cl ₆ = (0.610/1) × 1.16 × 10 ⁵ = 70760 Pa

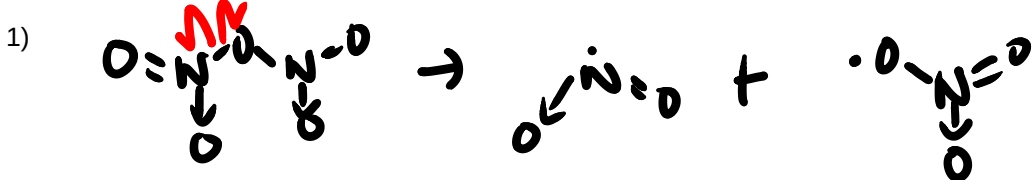
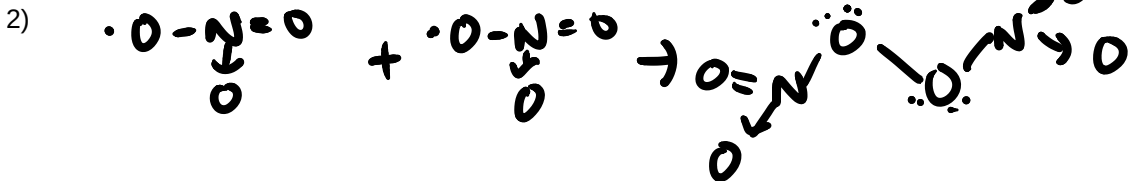
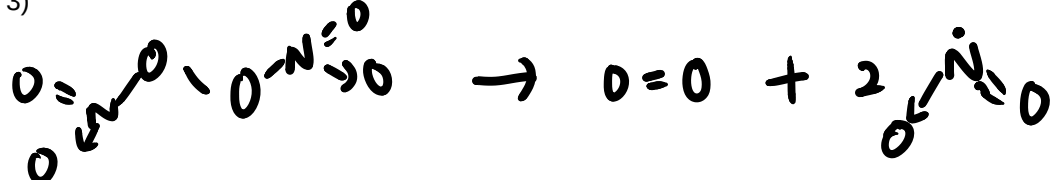
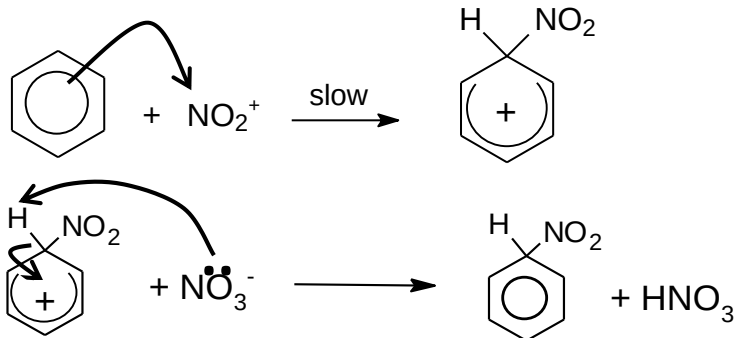
	Partial Pressure of $\text{AlCl}_3 = (0.390/1) \times 1.16 \times 10^5 = 45240 \text{ Pa}$
1(b)(vii)	$K_p = \frac{(P_{\text{AlCl}_3})^2}{P_{\text{Al}_2\text{Cl}_6}} = (45240)^2 / (70760) = 2.89 \times 10^4 \text{ Pa}$
1(c)(i)	<chem>CC(=O)c1ccccc1</chem> A: <chem>O=C1Cc2ccccc2C1</chem> B:
1(c)(ii)	Test: Add aqueous sodium hydroxide, aqueous iodine and warm. Observation: Decolourisation of brown solution and yellow ppt of CHI_3 is seen for A. Brown solution remains for B. CHI_3 and <chem>[O-]C(=O)c1ccccc1.[Na+]</chem>

2(a)(i)	1 mol dm⁻³ of I^- (aq) and 1 mol dm⁻³ of I_3^- (aq) Note: draw only one electrochemical cell.
2(a)(ii)	When the half cell of Cl_2/Cl^- is connected to the I_2/I^- From Data Booklet $2\text{e}^- + \text{Cl}_2 \rightleftharpoons 2\text{Cl}^- \quad +1.36 \text{ V} \quad \text{(Reduction)}$ $2\text{e}^- + \text{I}_2 \rightleftharpoons 2\text{I}^- \quad +0.54 \text{ V} \quad \text{(Oxidation)}$ Overall E_{cell} value = $+1.36 - 0.54 \text{ V} = \underline{+0.82 \text{ V}}$ will be register at the voltmeter. Since the $E^\circ (\text{Cl}_2 / \text{Cl}^-)$ is more positive than $E^\circ (\text{I}_2/\text{I}^-)$, Cl_2 is a stronger oxidising agent than I_2.
2(a)(iii)	In the presence of an external circuit, a feasible reaction would occur. $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$
2(b)(i)	$2\text{Fe}^{3+} + 2\text{I}^- \rightleftharpoons 2\text{Fe}^{2+} + \text{I}_2$ Note: a reversible arrow is used because an equilibrium was set up.
2(b)(ii)	When CN^- is added, it can form complexes with Fe^{2+} and Fe^{3+}, resulting in both $[\text{Fe}^{2+}]$ and $[\text{Fe}^{3+}]$ to decrease. $\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+} \quad +0.77 \text{ V}$ $[\text{Fe}(\text{CN})_6]^{3-} + \text{e}^- \rightleftharpoons [\text{Fe}(\text{CN})_6]^{4-} \quad +0.36 \text{ V}$ $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^- \quad +0.54 \text{ V}$ As seen from the electrode potential value, $[\text{Fe}(\text{CN})_6]^{3-}$ ($E^\circ = +0.36 \text{ V}$) is a weaker oxidising agent compared to I_2 ($E^\circ = +0.54 \text{ V}$) Thus, iodine will now prefer to undergo reduction to form I^- and $[\text{Fe}(\text{CN})_6]^{4-}$ will undergo oxidation. Therefore, <u>position of equilibrium in (i) will shift left</u> due to a <u>decrease in oxidising power of the complexed Fe(III).</u>

2(c)	For $\text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_2\text{H}_5\text{I}$ Bond length of $\text{C-Cl} < \text{C-I}$ Bond energy $\text{C-Cl} > \text{C-I}$ Ease of cleavage of $\text{C-Cl} < \text{C-I}$ Rate of hydrolysis $\text{C}_2\text{H}_5\text{Cl} < \text{C}_2\text{H}_5\text{I}$
2(d)(i)	 C: (Since only 1 mol of OH^- is given, the C-I bond will cleave first)  D:
2d(ii)	 E: Note: amines, being more basic, is a stronger nucleophile than alcohols.  F: (Step by Step guide of how F is formed, not required by question but for your understanding)  
2(e)(i)	Presence of dilute HCl (aq), reflux for several hours Or dilute NaOH (aq), reflux for several hours

2(e)(ii)	For phenylbutazone	
	dilute HCl (aq), heat	dilute NaOH (aq), heat
		
	For phenobarbital	
	dilute HCl (aq), heat	dilute NaOH (aq), heat
	 NH₄⁺ CO₂	 NH₃ CO₃²⁻

3(a)(i)	$\text{Mg}(\text{NO}_3)_2 \rightarrow \text{MgO} + 2\text{NO}_2 + \frac{1}{2} \text{O}_2$
3(a)(ii)	• Cationic radius of M^{2+} increases. • Charge remains constant. • Polarising power <u>of</u> cation decreases • polarising effect <u>on</u> anion decreases • Hence, stability of nitrates increases. Charge density of cation decreases. and M^{2+} is less able to distort the electron cloud of anion
3(b)(i)	
3(b)(ii)	$\text{N}_2\text{O}_5 \rightarrow 2\text{NO}_2 + \frac{1}{2} \text{O}_2$

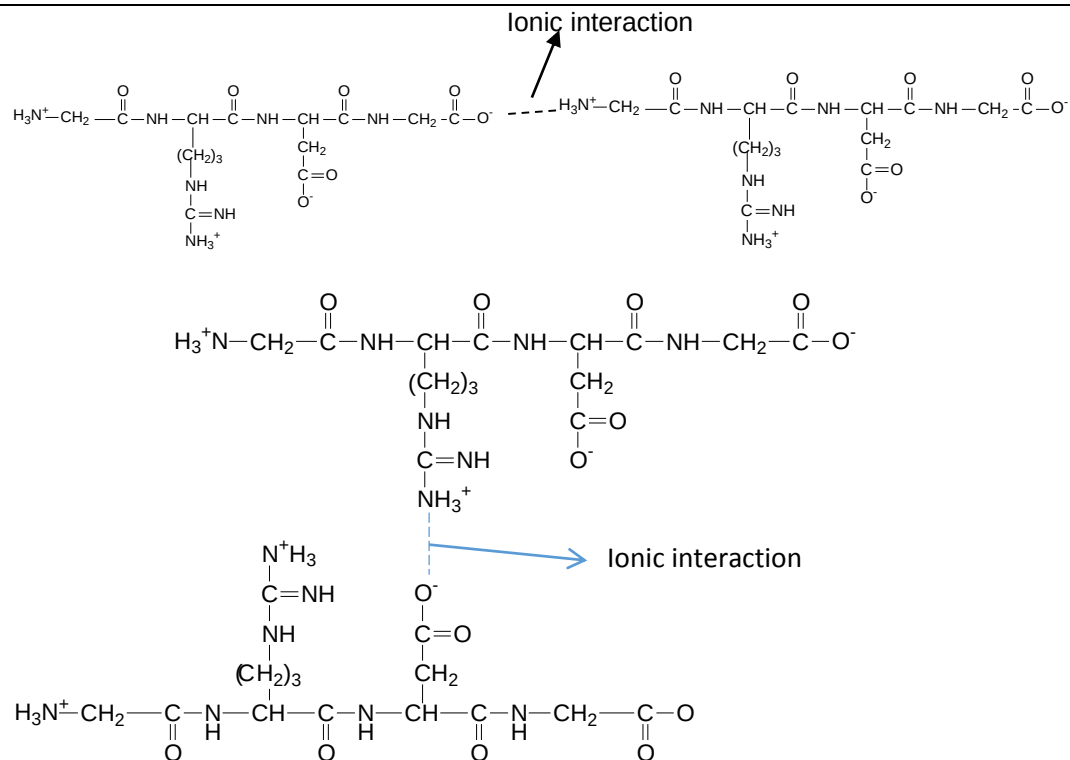
3(b)(iii)	1)  2)  3) 
3(b)(iv)	Step 1 is the slowest as it is given that the reaction is first order wrt N ₂ O ₅ , one N ₂ O ₅ molecule in step 1
3(c)(i)	HNO ₃ (NO ₃ ⁻ acts as base to remove H ⁺ from intermediate carbocation)
3(c)(ii)	N₂O₅ → NO₂⁺ + NO₃⁻ 
3(d)(i)	$\Delta H_{\text{rxn}} = \Sigma \Delta H_f(\text{prdt}) - \Sigma \Delta H_f(\text{rxtn}) = \Delta H_f(\text{NO}_3) + \Delta H_f(\text{NO}_2) - \Delta H_f(\text{N}_2\text{O}_5)$ $92.5 = \Delta H_f(\text{NO}_3) + 33.2 - 5$ $\Delta H_f(\text{NO}_3) = 64.3 \text{ kJ mol}^{-1}$
3(d)(ii)	$\Delta H_{\text{rxn}} = 2\Delta H_f(\text{NO}_2) - 2\Delta H_f(\text{NO}_3)$ $= 2 \times 33.2 - 2 \times 64.3$ $= -62.2 \text{ kJ mol}^{-1}$
3(e)	rxn I and II: $\Delta S > 0$ more disorderly, increase in amt of gases $\Delta G = \Delta H - T\Delta S$ Rxn I: $\Delta G = +ve - (+ve) = +ve$ (ΔG will only become $-ve$ at high temp) Rxn II: $\Delta G = -ve - (+ve) = \text{always } -ve$ Rxn II more spontaneous.
4(a)	e/m of proton: $1/1 = 1$ angle: 15° e/m of D⁺: $1/2$ → angle: 7.5° (opposite in direction to proton) e/m of T⁺: $1/3$ → angle: 5° e/m of He²⁺: $2/4 = 1/2$ → angle: 7.5°
4(b)(i)	to have angle of 5° , e/m of R is $1/3$. Since mass of R is 12, charge is $4+$.
4(b)(ii)	R has 6 protons. To have mass of 12, R has 6 neutrons. To have charge of $4+$, R has 2 electrons.
4(c)(i)	Number of electrons for Na⁺ and Si⁴⁺, P³⁻ and Cl⁻ are the same. (They are isoelectronic species). Anions P³⁻ and Cl⁻ are bigger than Na⁺ and Si⁴⁺ as they have an extra shell of electrons - Across the period, number of protons increases hence causing an increase in nuclear charge. Electrostatic forces of attraction between the nucleus and electrons increases thus pulling the electrons closer to the nucleus. - This explains why Si⁴⁺ ionic radius is smaller than Na⁺ and ionic radius of Cl⁻ is lesser than P³⁻

4(c)(ii)

Melting point: $\text{Si} > \text{Na} > \text{P}_4 > \text{Cl}_2$

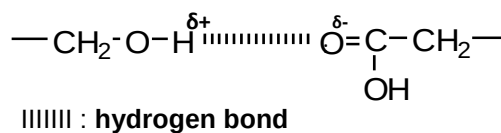
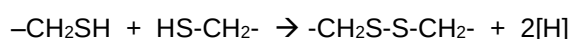
- **Si has a very high melting point**
 - Si has a giant molecular structure.
 - Very large amount of energy required to overcome the extensive covalent bonds between atoms in the three dimensional network structure.
 - **Metals have high melting points**
 - Metals have giant metallic structures.
 - Large amount of energy is required to overcome strong metallic bond.
- Since mp of Na < mp of Si → metallic bonding in sodium is weaker than covalent bonding in Si
- **P_4 , Cl_2 have low melting points**
 - P_4 , Cl_2 have simple molecular structures.
 - Small amount of energy is required to overcome the weak intermolecular van der Waals' forces of attraction.
 - **Melting point of $\text{P}_4 > \text{Cl}_2$**
 - Number of electrons of $\text{P}_4 > \text{Cl}_2$
 - Hence, the extensiveness of intermolecular van der Waals' forces of attraction of $\text{P}_4 > \text{Cl}_2$
 - The amount of energy required to overcome the stronger intermolecular van der Waals forces of $\text{P}_4 > \text{Cl}_2$

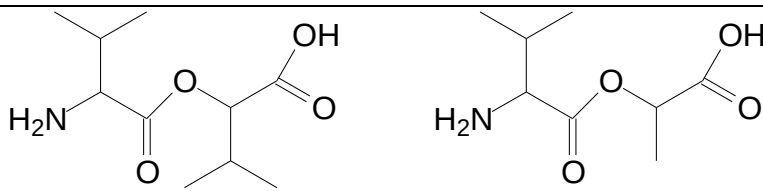
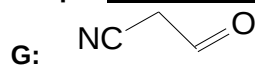
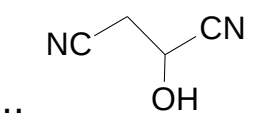
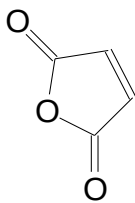
4(d)(i)

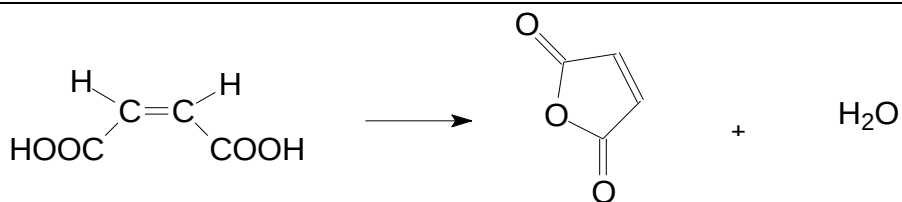


4(d)(ii)

Any 2 of the following

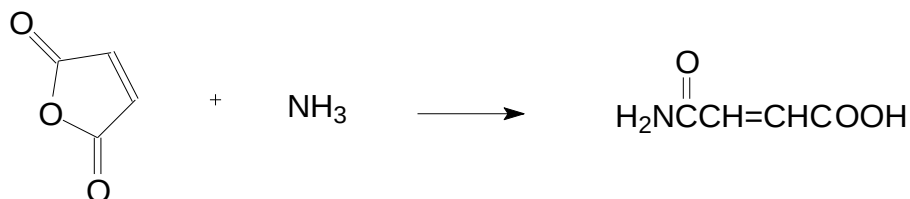
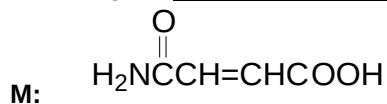
① hydrogen bonds between hydrophilic (polar) groups② van der Waals' interaction between non polar side chains③ disulfide linkages (S-S single bond) is formed when the -SH side chain of two cysteine residues undergo oxidation:

4(e)(i)	Ester
4(e)(ii)	
5(a)	(a) Step I: <u>alcoholic KCN, heat</u> G:  Step II: <u>HCN in NaOH(aq) / HCN in NaCN (aq), 10 – 20 °C</u> H:  Step III: <u>aqueous HCl / H2SO4, heat</u>
5(b)(i)	 Cis-trans isomerism
5(b)(ii)	excess concentrated sulfuric acid, heat at 170°C OR Passing alcohol vapour over alumina (Al₂O₃) catalyst, 350°C OR Phosphoric acid (H₃PO₄), heat at 200°C
5(b)(iii)	HOOCCH=CHCOOH $\rightleftharpoons$ HOOCCH=CHCOO⁻ + H⁺ ---- (1) Isomer K produces the more stable mono-anion on treatment with 1 mol of NaOH. This is because the pK_{a1} of isomer K is smaller. A smaller pK_{a1} value corresponds to a larger K_a value, indicating that the position of equilibrium (1) lies more to the right. This means that isomer K produces a more stable conjugate base (ie. The mono-anion) and is a stronger acid compared to isomer J.
5(b)(iv)	Displayed formula  The mono-anion produced by isomer K (cis isomer) can be stabilised by hydrogen bonding. This is due to the close proximity of the -COOH and -COO⁻ groups (they are on the same side in a cis-isomer). The -COOH and -COO⁻ groups are too far away from each other in the trans isomer and hence the mono-anion cannot be stabilised by hydrogen bonding.
5(c)	- L is a neutral compound => L is not a carboxylic acid - L does not undergo redox with Na metal => L does not contain a -OH group - L does not undergo condensation with 2,4-DNPH => L is not a carbonyl compound. - On heating J or K, 1 mol of J or K loses 1 mol of H₂O to form L. Hence, L is  Hence, isomer K produces L as both the -COOH groups must be on the same side to form L.

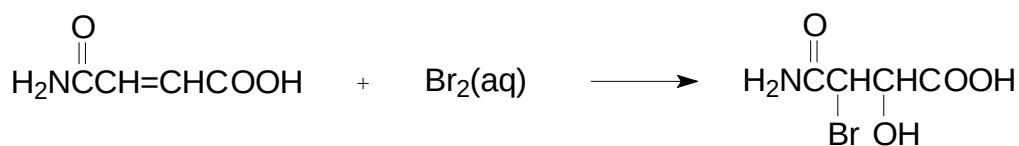


Eqn:

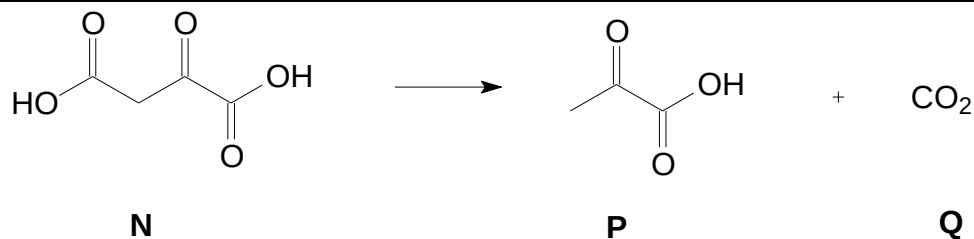
- **M** undergoes neutralization with NaOH but not with HCl => **M** is a carboxylic acid and is not an amine => **M** is an amide
- **M** undergoes electrophilic addition with Br₂ (aq) => **M** is an alkene



Eqns:



5(d)



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